10 The table shows information about the times taken by 100 people in a race.

Time (t minutes)	Frequency
$30 < t \leqslant 40$	10
$40 < t \leqslant 50$	22
$50 < t \leqslant 60$	28
$60 < t \leqslant 70$	20
$70 < t \leqslant 80$	15
$80 < t \leqslant 90$	5

(a) Complete the cumulative frequency table for this information.

Time (t minutes)	Cumulative frequency
$30 < t \leqslant 40$	10
$30 < t \leqslant 50$	32
$30 < t \leqslant 60$	60
$30 < t \leqslant 70$	80
$30 < t \leqslant 80$	95
$30 < t \leqslant 90$	100

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

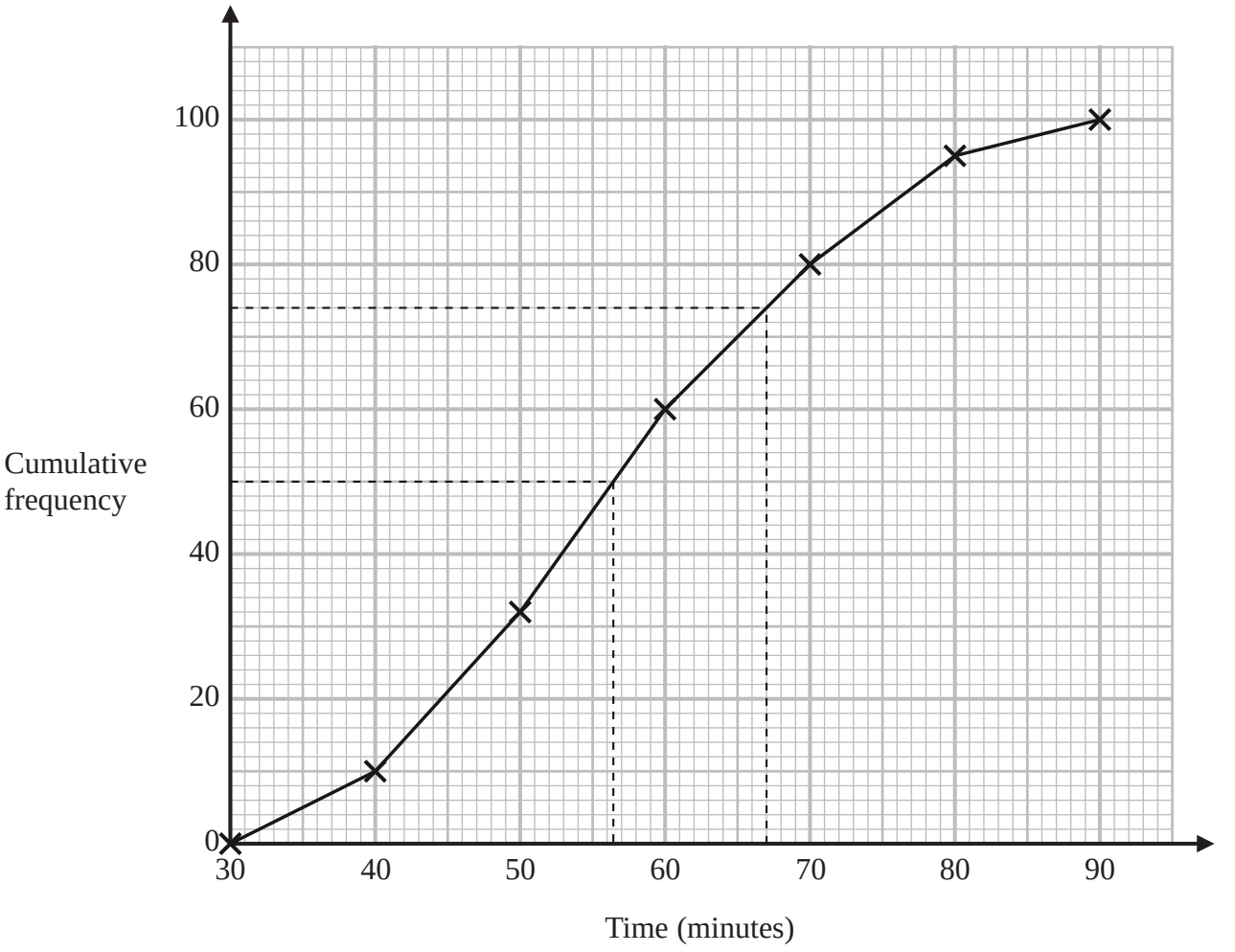
(2)

1 mark
Running totals: 10; +22; +28; +20; +15; +5.

2 marks total
Plot cumulative totals at UPPER class boundaries;
include the starting point (30, 0). Join in increasing order.

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(c) Use your graph to find an estimate for the median time.

50th value → about 56.4 minutes.

56.4 minutes

1 mark

Half of100 is50; read across then down.

(1)

(d) Use your graph to find an estimate for the number of people in the race who took longer than 67 minutes.

At67 minutes, cumulative frequency ≈74.

Longer than67: 100 – 74 = 26 people.

26

2 marks total

Read the number up to67, then subtract from100.

(2)

(Total for Question 10 is 6 marks)